

Intense but Self-Limiting: Neuroticism Composition and the Diffusion of Negative Sentiment in Korean Stock Message Boards

Jaeho Choi
Sogang University, Department of Economics

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Abstract

We study whether the composition of investor language predicts the diffusion of negative sentiment in online stock discussions. Using message-level sentiment and Big Five labels aggregated to daily board-level panels for Korean equities, we examine whether daily Neuroticism share forecasts next-day negativity after controlling for stock and date effects and standard discussion-state covariates. We find that higher daily Neuroticism share is associated with lower subsequent diffusion of negative sentiment, even though it is strongly positively associated with same-day negativity. This pattern is robust across board-level and panel specifications and is strongest in comparisons that hold current negativity approximately fixed. The evidence supports a reduced-form interpretation in which high-Neuroticism-composition days are intense but self-limiting discussion states rather than socially expansive ones. Our main contribution is to distinguish sentiment intensity from sentiment diffusion and to show that daily linguistic composition contains economically informative variation for that distinction.

Keywords: investor sentiment; online discussion boards; social diffusion; text analysis; Korea
JEL codes: G14; G41; D83; C23

1 Introduction

How negative sentiment spreads through investor communities is central to understanding retail market dynamics. Existing text-based finance research has established that online discussion contains information, but it has focused mainly on whether text predicts prices, volatility, or fundamentals (Tumarkin and Whitelaw, 2001; Antweiler and Frank, 2004; Das and Chen, 2007; Tetlock, 2007; Tetlock, Saar-Tsechansky, and Macskassy, 2008). Much less is known about whether different linguistic states imply different *propagation dynamics* of sentiment itself.

This paper studies that propagation margin in Korean stock message boards. The data combine message-level sentiment labels and Big Five personality labels, aggregated to daily board-level

measures. Our focal state variable is the daily share of Neuroticism-labeled language (daily Neuroticism share). We ask whether this composition measure predicts next-day diffusion of negative sentiment, not merely same-day negativity.

The distinction matters because standard sentiment measures can confound *intensity* (how negative discussion is today) with *diffusion* (whether negativity carries forward). A day can be highly negative yet socially self-limiting, or only moderately negative yet highly transmissive. Treating both as the same object obscures economically relevant heterogeneity in discussion states.

Empirically, three results organize the paper. First, board-level reduced-form diffusion fits show stable sign patterns: a higher daily Neuroticism share is associated with weaker next-day negative diffusion in a clear majority of boards with valid time-series variation. Second, panel estimates with stock and date fixed effects yield the same directional conclusion, including conservative specifications that control for full Big Five composition and use two-way clustered standard errors. Third, the relationship is strongest in comparisons that hold current negativity approximately fixed (stock-specific bins and trajectory controls), suggesting that the result is not a mechanical reflection of current tone levels.

At the same time, daily Neuroticism share is strongly positively associated with same-day negative share. This sign reversal across horizons is central to interpretation: high-Neuroticism-composition days appear to be *intense but self-limiting* discussion states. We therefore adopt a reduced-form reading and do not interpret this measure as a structural contagion parameter.

The paper contributes in three ways. First, it provides a measurement framework that separates sentiment intensity from sentiment diffusion. Second, it shows that daily linguistic composition adds incremental information about that distinction beyond scalar negativity measures. Third, it provides evidence from a non-U.S. retail-investor setting where message boards remain an active channel of market discourse.

Our identification claims are intentionally modest. The results characterize robust state dependence in discussion dynamics, but they do not by themselves isolate a unique transmission mechanism or imply a universal pricing effect. Interpreted this way, the evidence supports a tractable and economically useful distinction between being negative and becoming diffusively negative.

2 Related Literature and Contribution

This paper relates to four strands of literature and contributes at their intersection.

First, it connects to textual-finance studies showing that investor communication and media language contain information about market outcomes (Tumarkin and Whitelaw, 2001; Antweiler and Frank, 2004; Das and Chen, 2007; Tetlock, 2007; Tetlock, Saar-Tsechansky, and Macskassy, 2008). The central contribution of that literature is to establish that text predicts returns, volatility, or firm-level fundamentals. Relative to this work, we shift the dependent variable from market

prices to *discussion dynamics*: whether negative sentiment propagates across days within investor communities.

Second, it complements research on attention and social transmission in financial markets (Hong, Kubik, and Stein, 2004; Barber and Odean, 2008; Da, Engelberg, and Gao, 2011; Boulland, Degeorge, and Ginglinger, 2017; Rantala, 2019; Hirshleifer, Peng, and Wang, 2025; Sui and Wang, 2025). That literature emphasizes salience, social interaction, and network-mediated diffusion of investment ideas. Our design is complementary rather than network-structural: we do not observe explicit follower links, but use repeated board-level states to identify whether some negative discussion states are more diffusively persistent than others.

Third, it is related to recent evidence that social-media attention and investor opinion disagreement affect trading behavior and information incorporation (Wu, Zhao, and Huang, 2024; Ton et al., 2024; Zhao, Liu, and Chen, 2025). This strand highlights heterogeneity in beliefs and attention shocks as key drivers of market dynamics. Our results add a distinct composition margin: even conditional on current negativity, daily language composition predicts next-day diffusion.

Fourth, it relates to language-based measures of psychology and personality (Tausczik and Pennebaker, 2010; Hrazdil et al., 2020; Jiang, Peng, and Yan, 2024). We use this literature in a deliberately reduced-form way. We do not infer stable investor-level traits; instead, we treat daily Neuroticism share as a board-state variable that captures discussion composition at a given point in time.

Our contribution is both conceptual and empirical. Conceptually, we distinguish sentiment *intensity* from sentiment *diffusion*, which scalar negativity measures often conflate. Empirically, we show that this distinction is informative in a non-U.S. retail-investor setting and robust to fixed-effects, within-bin, and trajectory-controlled comparisons.

3 Data and Variable Construction

3.1 Sample construction

The unit of observation is the stock-day. The baseline panel contains 36,881 stock-days from 99 stock boards over 942 trading dates. The stock-level reduced-form diffusion exercise is estimated separately for 95 boards with enough valid consecutive observations. Board posts are assigned to trading days using a 15:30 cut-off, so that post-close messages are shifted to the next trading day. The same alignment rule is used in the intraday and disclosure-control exercises.

The underlying message dataset contains post-level labels for sentiment and Big Five categories. In the source labeling pipeline, these two label families are generated with OpenAI GPT-4o under fixed annotation prompts and then mapped to daily board-level aggregates. Using LLMs for text annotation is increasingly supported by validation evidence showing high agreement and strong performance relative to crowd and expert coders in comparable classification tasks (Gilardi, Alizadeh,

and Kubli, 2023; Törnberg, 2025; Alizadeh et al., 2025). We take these labels as inputs and aggregate them to the stock-day level. Let $M_{i,t}$ denote the number of labeled messages for stock i on day t . The main outcome is the daily negative-share measure

$$Neg_{i,t} = \frac{\text{negative messages}_{i,t}}{M_{i,t}}.$$

The focal regressor is

$$BN_{i,t} = \frac{\text{Neuroticism-labeled messages}_{i,t}}{M_{i,t}},$$

which corresponds to the variable `bN` in the dataset. The paper also uses the daily shares of Agreeableness, Conscientiousness, Extraversion, and Openness, denoted `bA`, `bC`, `bE`, and `bO`. Because these shares are compositional and sum to one, specifications that include the full Big Five vector omit one reference category; throughout the conservative daily-panel exercises, Agreeableness is the omitted share.

To reduce measurement noise in share outcomes, the daily-panel sample keeps stock-days with at least ten messages. Attention is measured as $\log(1 + M_{i,t})$. Same-day market controls include raw return, absolute return, and realized volatility, together with one-day lags of return and volatility. For the disclosure-control exercise, the paper merges Korean Exchange (KRX KIND) disclosure information at the stock-day level.

3.2 Additional variables

Several derived variables are useful for interpretation. The variable `bN_neg` measures the share of Neuroticism-labeled messages among *negative* messages only. This is closer to a “style of negativity” measure than raw `bN`, which is computed over all messages. The variable `clr_bN` is the centered-log-ratio transform of daily Neuroticism share within the Big Five composition. The paper also constructs stock-specific bins of current negativity and uses them for within-bin comparisons that hold the current level of negativity approximately fixed.

Table 1 reports summary statistics for the baseline next-day-regression sample. The average stock-day contains 87 messages, the mean current negative share is 0.390, and the mean daily `bN` share is 0.448. The distribution is wide enough to support both cross-sectional and within-stock variation.

Table 1: Summary statistics: baseline daily-panel sample

Variable	Mean	Std. dev.	P25	Median	P75
Messages per day (M)	87.0042	173.2467	20.0000	40.0000	89.0000
Current negative share (Neg_t)	0.3902	0.1157	0.3143	0.3919	0.4686
Next-day negative share (Neg_{t+1})	0.3884	0.1172	0.3125	0.3916	0.4681
Daily Neuroticism share (bN)	0.4481	0.1104	0.3793	0.4553	0.5263
Daily Conscientiousness share	0.1478	0.0813	0.0833	0.1379	0.1944
Daily Openness share	0.1851	0.0745	0.1356	0.1818	0.2286
Attention: $\log(1 + M_t)$	3.8572	1.0003	3.0445	3.7136	4.4998
Return	-0.0003	0.0293	-0.0144	0.0000	0.0122
Absolute return	0.0196	0.0218	0.0055	0.0134	0.0264
Realized volatility	0.0009	0.0030	0.0000	0.0002	0.0007
Lagged return	-0.0002	0.0288	-0.0140	0.0000	0.0118
Lagged realized volatility	0.0008	0.0030	0.0000	0.0002	0.0007

Notes: The table reports descriptive statistics for the 36,881 stock-days used in the baseline next-day-share regression. The sample includes 99 stocks and 942 trading dates.

4 Empirical Design

4.1 Reduced-form diffusion equation

The stock-level exercise uses a parsimonious diffusion equation to summarize nonlinear propagation in negative-message share:

$$n_{i,t+1} = n_{i,t} + \beta_{i,t}n_{i,t}(1 - n_{i,t}) - \gamma_in_{i,t}, \quad (1)$$

with

$$\log(\beta_{i,t}) = \beta_{0,i} + \theta'X_{i,t}. \quad (2)$$

Here $n_{i,t}$ is daily negative share rather than a literal infection rate. Silent readers are unobserved, the active set of writers changes over time, and the susceptible population is not fixed. Accordingly, Equations (1)–(2) are interpreted as a compact reduced-form description of diffusion rather than as a structural epidemic model. The stock-level coefficients are used primarily for sign stability, not for structural calibration.

4.2 Daily-panel regressions

The main daily-panel equation relates next-day negativity to current-day language composition:

$$Neg_{i,t+1} = \alpha_i + \lambda_t + \rho Neg_{i,t} + \theta BN_{i,t} + \eta(BN_{i,t} \times Neg_{i,t}) + \Gamma'Z_{i,t} + \varepsilon_{i,t+1}, \quad (3)$$

where α_i and λ_t are stock and date fixed effects, respectively, and $Z_{i,t}$ contains the remaining Big Five shares, attention, same-day return, same-day absolute return, same-day realized volatility, lagged return, and lagged volatility. Regressions are weighted by next-day message count so that

stock-days with more precisely measured shares receive more weight.

Two design choices are central. First, the paper reports a conservative reference-coded specification in which the full Big Five composition is included and standard errors are clustered two ways by stock and date. Second, the paper uses within-bin comparisons that condition on stock-specific bins of current negativity. This design asks whether, among days with similar current negativity, higher **bN** predicts more or less next-day diffusion. Follow-up specifications add explicit trajectory controls—the current change in negativity and its lag—and disclosure-day controls from KRX KIND.

The regressions are descriptive rather than causal. The identifying variation comes from within-stock changes in board composition after removing common date effects and controlling for current negativity and market conditions. The paper therefore interprets θ as a reduced-form discussion-state coefficient.

5 Main Results

5.1 Stock-level sign stability

Table 2 summarizes the board-by-board diffusion fits. In the reference-coded stock-level exercise, the loading on **bN** is negative on average and at the median, and significantly negative in 20 boards versus significantly positive in only 2. The corresponding baseline specification yields an even more negative mean loading. The coefficient magnitudes are not directly structural because they come from nonlinear stock-specific fits, but the sign pattern is informative. **bN** is the most consistently negative personality loading in the diffusion equation.

Table 2: Stock-level reduced-form diffusion summaries

Specification	Mean loading	Median loading	Sig. negative	Sig. positive
Baseline stock-level diffusion fit: bN	-1.2646	-0.5475	28	3
Reference-coded diffusion fit: bN	-0.6248	-0.7029	20	2

Notes: The baseline row summarizes 95 successful board-specific reduced-form diffusion fits that include **bN**, **bC**, **bO**, attention, lagged return, and lagged volatility in the diffusion equation. The reference-coded row omits Agreeableness from the Big Five composition to respect the adding-up constraint. The table is meant to show sign stability rather than structural effect size.

5.2 Daily-panel evidence on future diffusion

Table 3 reports the main daily-panel estimates. Column (1) is the baseline count-weighted specification with stock-clustered standard errors. The coefficient on **bN** is -0.1535 and statistically significant at conventional levels. Column (2) adds the full reference-coded Big Five composition and clusters standard errors two ways by stock and date; the coefficient remains negative at -0.1037 but becomes less precise. Columns (3) and (4) focus on within-bin comparisons. Once current negativity

is held approximately fixed by stock-specific bins, the coefficient on **bN** is clearly negative. It remains negative and statistically significant after adding trajectory controls and disclosure controls.

These estimates are the paper’s core result. Higher daily **bN** composition predicts *less* subsequent diffusion of negative sentiment. The strongest version of the result is not the fully parametric specification in column (2), but the within-bin comparisons in columns (3) and (4), where the level of current negativity is more tightly controlled.

Table 3: Daily-panel evidence on future negativity

	(1) Baseline FE	(2) Full composition	(3) Within-bin	(4) Within-bin + trajectory
Daily bN	-0.1535** (0.0632)	-0.1037* (0.0594)	-0.0874*** (0.0281)	-0.0641** (0.0251)
bN × current negativity	0.2052* (0.1074)	0.2059* (0.1083)		
Current-negativity bins	No	No	Yes	Yes
Trajectory controls	No	No	No	Yes
Disclosure controls	No	No	No	Yes
Other Big Five controls	Partial (bC, bO)	Full (A omitted)	Full (A omitted)	Full (A omitted)
Stock fixed effects	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes
Clustering	Stock	Stock + date	Stock + date	Stock + date
Observations	36,881	36,881	36,877	36,698
Stocks	99	99	98	98

Notes: The dependent variable in columns (1) and (2) is next-day negative share, $Neg_{i,t+1}$. In columns (3) and (4), the dependent variable is the next-day change in negative share within stock-specific bins of current negativity. All columns are weighted by next-day message count. Standard errors are in parentheses. "Partial (bC, bO)" includes only Conscientiousness and Openness shares. "Full (A omitted)" includes the full Big Five composition with Agreeableness omitted as the reference category due to the adding-up constraint. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 1 visualizes the same progression across specifications. The sign is uniformly negative. Figure 2 provides a nonparametric summary: moving from the lowest to the highest **bN** quintile, the average next-day change in negative share declines from strongly positive to strongly negative, and the model-implied diffusion proxy declines as well.

5.3 Intensity versus diffusion

A natural concern is that **bN** is simply another way of measuring current negativity. That concern is partly correct. In a regression of current-day negative share on current-day **bN**, the coefficient is 0.7671 with a p -value below 10^{-40} . High-**bN** days are therefore unmistakably high-negativity days.

But this is exactly why the distinction between *level* and *diffusion* matters. If **bN** were merely a synonym for contagious pessimism, then conditioning on current negativity should remove the negative future-diffusion effect. Instead, the within-bin estimates remain negative and statistically significant. The evidence points to an intense but self-limiting state: a day can be highly negative in the present while still being less likely to generate additional negativity tomorrow.

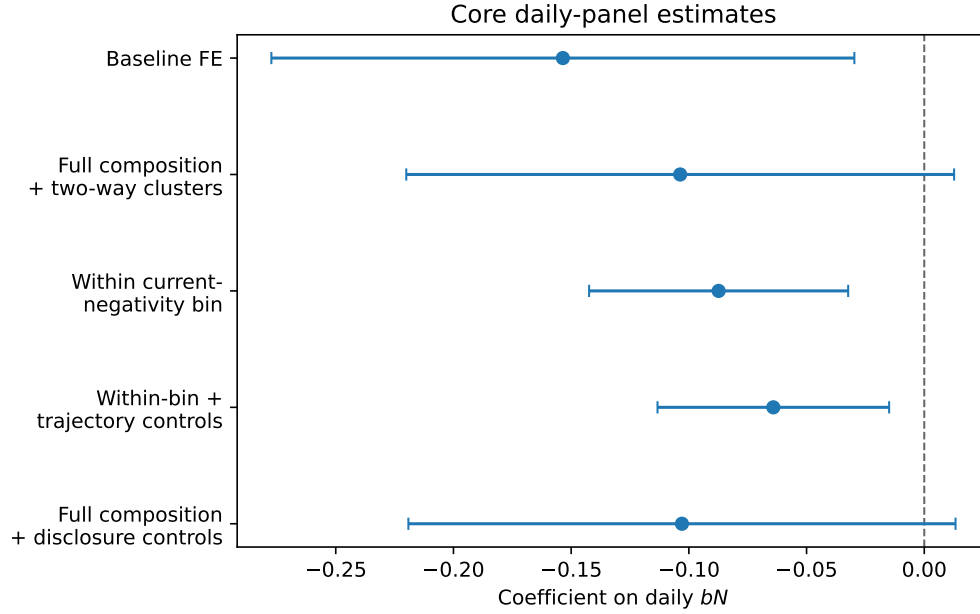


Figure 1: Core daily-panel estimates

Notes: Points show coefficient estimates on daily bN ; horizontal bars are 95% confidence intervals. The figure summarizes the progression from the baseline fixed-effects regression to the conservative and within-bin specifications.

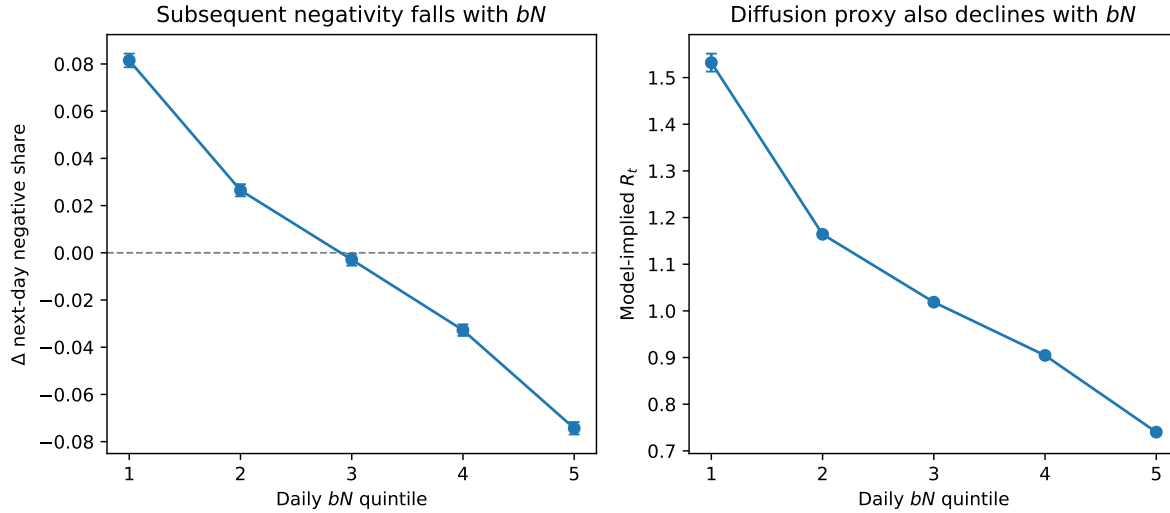


Figure 2: Diffusion metrics by daily bN quintile

Notes: The left panel plots the mean next-day change in negative share by daily bN quintile with 95% confidence intervals. The right panel plots the corresponding model-implied diffusion proxy. Both measures decline monotonically with bN .

6 Interpretation and Robustness

Table 4 collects the main interpretive exercises. Panel A contrasts the contemporaneous and future-diffusion roles of **bN**. The first row shows the strong positive same-day relation between **bN** and current negativity. The second row shows that **bN_neg**—Neuroticism among *negative* posts only—has a positive coefficient in the next-day regression. The third row uses the centered-log-ratio transformation of Neuroticism share and yields a negative but imprecise coefficient. Together, these results suggest that the main finding is not driven by neurotic wording *inside* negative posts alone.

Panel B adds disclosure controls and decomposition results. The full-composition coefficient on **bN** remains almost unchanged after KRX disclosure controls are added. An adverse disclosure-day indicator is itself negative, but it does not explain away the core **bN** coefficient. In the trajectory-controlled decomposition, the negative-post component is essentially zero, whereas the non-negative-post component is negative and statistically significant. The Wald test does not reject equality of the two coefficients, so this evidence should be read directionally rather than decisively. Still, the point estimates are more consistent with a broader board-composition interpretation than with a purely negative-post-language interpretation.

Table 4: Interpretation and robustness

Specification and focal regressor	Coefficient	S.E.	<i>p</i> -value
<i>Panel A. Current intensity versus future diffusion</i>			
Current negative share on current-day bN	0.7671	0.0301	$< 10^{-40}$
Next-day negative share on bN_neg	0.0994	0.0461	0.0337
Next-day negative share on clr_bN	-0.0164	0.0115	0.1580
Morning bN on afternoon negative share	0.0234	0.0302	0.4404
Morning bN_neg on afternoon negative share	0.0664	0.0214	0.0025
<i>Panel B. Disclosure controls and decomposition</i>			
Full-composition bN with disclosure controls	-0.1029	0.0593	0.0857
Adverse disclosure indicator	-0.0266	0.0113	0.0201
Negative-post component of daily bN	-0.0023	0.0490	0.9621
Non-negative-post component of daily bN	-0.0918	0.0370	0.0147
Difference: negative minus non-negative component	0.0895	0.0730	0.2233
Grouped-binomial coefficient on daily bN	-0.3533	0.2471	0.1527

Notes: Panel A contrasts current intensity and subsequent diffusion. The first row uses current-day negative share as the dependent variable. The intraday rows use morning-to-afternoon windows with stock and date fixed effects. Panel B reports the main disclosure-control and decomposition exercises. The grouped-binomial estimate is a nonlinear robustness check for the share outcome and is best interpreted directionally.

Appendix Table 5 shows that the conservative local-projection coefficients remain negative through horizons 2 and 3, although only horizon 1 is reasonably precise. This pattern reinforces the paper’s preferred interpretation: **bN** is primarily informative about near-horizon diffusion rather than about a long, persistent propagation process.

The paper also considered broader mechanism exercises on participation, herding, and market feedback. Those results are mixed and do not sharpen the central claim. For that reason, they are

treated as secondary diagnostics rather than as core evidence.

7 Conclusion

The paper documents a robust distinction between the *level* of negativity and the *diffusion* of negativity in online investor discussion. Daily Neuroticism composition is strongly positively associated with current-day negativity but negatively associated with subsequent diffusion of negative sentiment. The result survives stock-level sign-stability exercises, conservative fixed-effects regressions, within-bin comparisons, and trajectory-controlled follow-ups.

This finding changes the interpretation of board sentiment. A day with high negative tone is not necessarily a day on which negativity is most socially expansive. High-Neuroticism-composition days look more like intense but self-limiting episodes. The contribution is therefore methodological as much as behavioral: sentiment measurement should separate intensity from diffusion whenever the goal is to track the social spread of beliefs.

The evidence remains reduced form. The paper does not identify a structural contagion mechanism and it does not claim a universal market-pricing effect. What it does provide is a stable empirical characterization of a discussion state that standard scalar sentiment measures miss. That distinction should be useful for future work on retail attention, online social finance, and text-based measures of market sentiment.

A Additional Tables and Figure

A.1 LLM labeling prompt (sentiment and Big Five)

For reproducibility, the labeling pipeline uses the following base prompt template with each forum comment inserted at <COMMENT_TEXT>.

You are an expert in analyzing stock market discussions and psychological traits. The given content is a comment from a stock market forum. Your task is twofold: first, extract the psychological sentiment of this comment as Positive, Negative, or Neutral. Second, identify the most relevant Big Five personality trait (Openness to Experience, Conscientiousness, Extraversion, Agreeableness, Neuroticism) based on the language and sentiment expressed in the comment.

""<COMMENT_TEXT>""

- 1) Based on the overall sentiment and language used in the text, assign one of the sentiment labels (Positive, Negative, Neutral) that best represents the sentiment of the comment. Explain in detail why you chose this label by providing specific examples or phrases from the text that illustrate the sentiment.
- 2) Based on the sentiment and language style, assign one of the Big Five personality trait labels (Openness to Experience, Conscientiousness, Extraversion, Agreeableness, Neuroticism) that best represents the personality trait expressed in the comment. Provide specific examples or phrases from the text that support this classification.

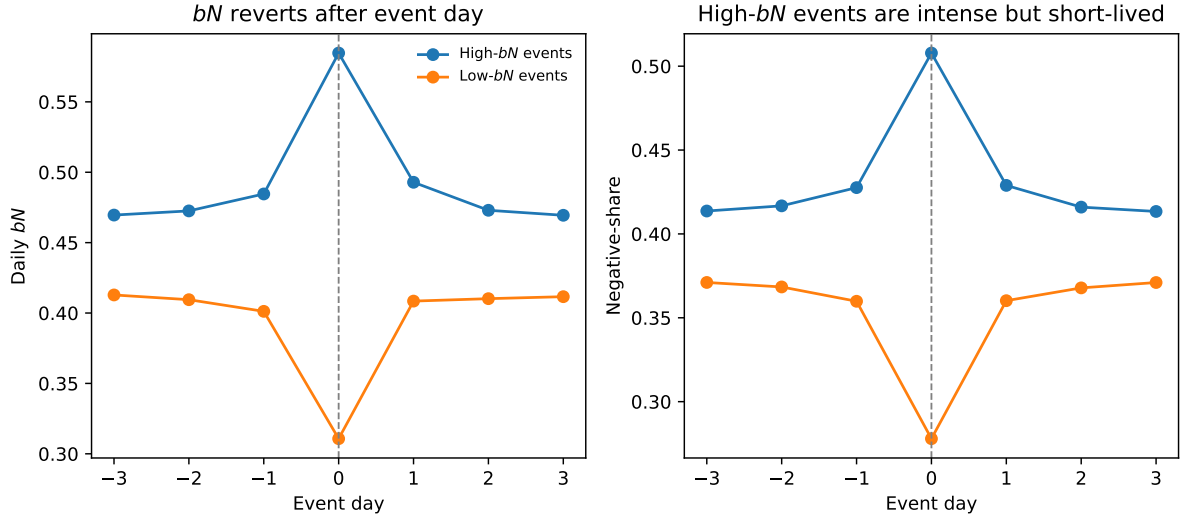
Your response must be in JSON format with the following keys:

- Sentiment Label (Positive, Negative, Neutral)
- Sentiment Explanation
- Big Five Label (Openness to Experience, Conscientiousness, Extraversion, Agreeableness, Neuroticism)
- Big Five Explanation

Table 5: Weighting and horizon robustness

Specification and focal regressor	Coefficient	S.E.	<i>p</i> -value
Full-composition FE, unweighted	-0.0489	0.0320	0.1295
Full-composition FE, $\sqrt{M_{t+1}}$ weights	-0.0795	0.0472	0.0951
Full-composition FE, $M_t \geq 30$	-0.0706	0.0680	0.3022
Local projection, horizon 1	-0.1037	0.0594	0.0839
Local projection, horizon 2	-0.0681	0.0659	0.3039
Local projection, horizon 3	-0.1067	0.0713	0.1380

Notes: All rows are based on the conservative full-composition specification with stock and date fixed effects. The first three rows change the weighting scheme or message-count threshold. The last three rows report local projections for future negative share at horizons 1 through 3.

**Figure 3:** Event-time patterns around high-bN days

Notes: The figure compares event windows centered on high-bN and low-bN days. Both bN and negative share spike on the event day for high-bN events and largely revert afterward, consistent with an intense but short-lived discussion state.

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